1. **Quantum sensing I**

   Consider the sensing problem where we have a unitary $U_\phi = e^{-i\phi/2}|0\rangle\langle 0| + e^{i\phi/2}|1\rangle\langle 1|$, and wish to use a single quantum probe in the state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ in order to learn information about ϕ. In this question we will consider performing a measurement of the operator $Y = |y+\rangle\langle y+| - |y-\rangle\langle y-|$, where $|y+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ and $|y-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$.

   (a) Calculate the probabilities to obtain the outcomes $+1$ and -1 when measuring the state $|\psi_\phi\rangle = U_\phi|+\rangle$. Use these probabilities to determine the expectation value $\langle Y\rangle_\phi$.
   Hint: You may find it helpful to simplify your answer using the trigonometric identities $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$ *and* $\sin(\alpha+\beta) = \cos\alpha\sin\beta + \sin\alpha\cos\beta$.

   (b) Calculate $\langle Y\rangle_\phi$ directly using the formula $\langle Y\rangle_\phi = \langle\psi_\phi|Y|\psi_\phi\rangle$ and $Y = i|1\rangle\langle 0| - i|0\rangle\langle 1|$, and verify that you obtain the same answer as in part (a).

   (c) Show that $\Delta Y_\phi = |\cos\phi|$.

   (d) Assume that ϕ is close to the value ϕ_0, and the estimated expectation value is y_0 after N repetitions of the procedure. Calculate the sensitivity $\Delta\phi_0$ of this sensing protocol. How does it compare to the protocol based upon an X measurement from the notes?

2. **Quantum sensing II**

   Consider the sensing problem where we have a unitary $U_\phi = e^{-i\phi/2}|0\rangle\langle 0| + e^{i\phi/2}|1\rangle\langle 1|$, and wish to use a single quantum probe in the state $|\chi_\theta\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle$, with $0 \le \theta \le \pi$, in order to learn information about ϕ. In this question we we will consider performing a measurement of of the operator $X = |+\rangle\langle +| - |-\rangle\langle -|$, where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

   (a) Show that $\langle X\rangle_{\theta,\phi} = \sin\theta\cos\phi$, the expected value of the operator X when measured on the probe state after application of the unitary, $|\chi_{\theta,\phi}\rangle = U_\phi|\chi_\theta\rangle$.

   (b) Show that $\Delta X_{\theta,\phi} = |\sin\phi|\sqrt{1+\cos^2\theta\cot^2\phi}$ and that $\frac{\partial\langle X\rangle_{\theta,\phi}}{\partial\phi} = -\sin\theta\sin\phi$.

   (c) Assume that ϕ is close to the value ϕ_0, and the estimated expectation value is x_0 after N repetitions of the procedure. Calculate the sensitivity $\Delta\phi_0$ of this sensing protocol. How does it compare to the protocol based upon an X measurement from the notes when the probe state was $|+\rangle$? What happens when $\theta = 0$ or $\theta = \pi$. Does this make sense?

3. **Quantum sensing III**

   Consider the sensing problem where we have a unitary $U_\phi = e^{-i\phi/2}|0\rangle\langle 0| + e^{i\phi/2}|1\rangle\langle 1|$, and wish to use an n-qubit quantum probe in the state $|\chi_\theta^n\rangle = \cos\frac{\theta}{2}|0\rangle^{\otimes n} + \sin\frac{\theta}{2}|1\rangle^{\otimes n}$, with $0 \le \theta \le \pi$, in order to learn information about ϕ.

   (a) Show that we can prepare the state $|\chi_\theta^n\rangle$ by preparing $|\chi_\theta\rangle|0\rangle^{\otimes n-1}$, and successively applying CNOT unitary operators between the first and k^{th} qubit, for $2 \le k \le n$.

   (b) Find the state of the n-qubit probe state, after it has probed the unitary, i.e. calculate $|\chi_{\theta,\phi}^n\rangle = (U_\phi \otimes \cdots \otimes U_\phi)|\chi_\theta^n\rangle$.

   (c) By applying the sequence of CNOT unitary operators from part (a) in reverse order, show that we end up in the state $|\chi_{\theta,n\phi}\rangle|0\rangle^{\otimes n-1}$.

   (d) Consider measuring X on the first qubit. Show that now $\langle X\rangle_{\theta,\phi} = \sin\theta\cos n\phi$, and $\Delta X_{\theta,\phi} = |\sin n\phi|\sqrt{1+\cos^2\theta\cot^2 n\phi}$.

   (e) Assume that ϕ is close to ϕ_0, and the estimated expectation value is x_0 after repeating the n-probe procedure N times. Calculate the sensitivity $\Delta\phi_0$ of this sensing protocol. How does it compare to the protocol from question 3?